The Conservation Theorem for Algebraic Cycles: Classical Logic Is Scaffolding

Paper 102 of the Constructive Reverse Mathematics Series

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Abstract

We prove a strong form of the Conservation Conjecture (Paper 100, Conjecture 7.1) for its arithmetical scope: every CLASS theorem about algebraic cycles on smooth projective varieties whose conclusion is arithmetical of complexity $\leq \Pi^0_2$ admits a proof in pure BISH. The argument composes three results from mathematical logic. First, a *coding lemma* establishes that the standard algebraic cycle statements—algebraicity (Σ^0_1), L-value non-vanishing (Σ^0_1), rank equalities (Π^0_2), and Fermat-type universals (Π^0_1)—are arithmetical sentences of complexity $\leq \Pi^0_2$ over the natural numbers. Second, a *reduction lemma* shows that classical proofs in arithmetic geometry formalise in systems conservative over Peano Arithmetic for arithmetical sentences (McLarty 2010). Third, PA is Π^0_2-conservative over HA (Friedman 1978): Peano Arithmetic and Heyting Arithmetic have the same provably total recursive functions, so every Π^0_2 theorem of PA is already a theorem of HA—and hence of BISH—without Markov's Principle.

Combined with the Archimedean Obstruction Thesis (Paper 98, Theorem 5.1), this seals the programme's logical architecture: the logical cost of arithmetic geometry is the logical cost of $\mathbb{R}$—not the ring structure, but the completeness and total ordering at the Archimedean place. The lower bound says the continuum *forces* CLASS on every proof path through the Betti realization. The upper bound proved here says that for arithmetical conclusions, CLASS is always *eliminable*. The continuum is a removable mirror: it reflects algebraic truth but adds no algebraic content.

Five case studies from the programme (Papers 86, 96, 99, 100) verify consistency. In every case where an explicit constructive proof is known, it achieves pure BISH—matching the conservation guarantee exactly. One genuinely new prediction is made: the rank equality $\operatorname{rk} E(\mathbb{Q}) = 1$ for the curve 37a1 is provable in BISH, though no explicit constructive proof is currently known.

A companion 0-`sorry` Lean 4 bundle (925 lines, 6 files) mechanically verifies the CRM framework and all case-study consistency checks.

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1 Introduction

The Conservation Conjecture, stated in Paper 100 [25] as the programme's principal open problem, asks: if a theorem about algebraic cycles on smooth projective varieties is proved using CLASS but its conclusion is expressible in BISH, does a proof exist at a strictly lower CRM level? We prove the conjecture for its arithmetical scope—conclusions of complexity $\leq \Pi^0_2$—and show the target is pure BISH. The Coding Lemma establishes that the standard algebraic cycle statements all fall within this scope.

Across 101 papers of calibration data—17 detailed case studies spanning K3 surfaces, abelian varieties, elliptic curves, and Fermat's Last Theorem—zero counterexamples have been found. Every time a CLASS proof has a BISH-statable conclusion, an alternative proof at lower CRM level has been located. Paper 100 asked whether this is a law or a coincidence.

It is a law. The proof is not internal to arithmetic geometry; it comes from mathematical logic, specifically from the proof-theoretic tradition of Gödel, Gentzen, and Friedman.

Theorem 1.1 (Conservation for Algebraic Cycles)**.** *Let X be a smooth projective variety over $\overline{\mathbb{Q}}$, and let φ be an arithmetical statement of complexity $\leq \Pi^0_2$ about algebraic cycles on X. If φ is provable in ZFC (using any classical methods: Hodge theory, Betti realization, analytic continuation, etc.), then φ is provable in pure BISH.*

Remark 1.2. This is stronger than conservation to BISH + MP (the naïve reading of the Gödel–Gentzen negative translation). The strengthening comes from Friedman's 1978 result that PA and HA have the same provably total recursive functions, which gives Π^0_2-conservation of PA over HA without Markov's Principle. See §5 for details.

Theorem 1.3 (Coding Lemma)**.** *The standard algebraic cycle statements are arithmetical of complexity $\leq \Pi^0_2$:*

Statement type	***Complexity***	***Target CRM level***
"α is algebraic"	Σ^0_1	*BISH*
"$L(E,1) \neq 0$"	Σ^0_1	*BISH*
"$\operatorname{rk} E(\mathbb{Q}) = r$"	Π^0_2	*BISH*
"$x^n + y^n \neq z^n$ for all x, y, z, n"	Π^0_1	*BISH*

Remark 1.4 (Sha finiteness is outside scope)**.** The statement "Sha(E) is finite" has natural encoding $\exists N. \forall \phi \in \mathrm{Sha}(E). \mathrm{ord}(\phi) \leq N$, which is Σ^0_2 ($\exists\forall$ with decidable matrix), not Π^0_2. It therefore falls outside the scope of the Conservation Theorem. A Π^0_2 encoding of Sha finiteness would require reformulating the statement as a $\forall\exists$ sentence; no natural such reformulation is known.

1.1 CRM primer

The CRM hierarchy classifies mathematical theorems by the weakest logical principle sufficient to prove them:

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

BISH is Bishop's constructive mathematics [2]: intuitionistic logic with dependent choice. MP (Markov's Principle) asserts that if $\neg\neg\exists n. P(n)$ and P is decidable, then $\exists n. P(n)$. In the presence of countable choice (which BISH includes), MP is implied by LLPO and hence by all stronger omniscience principles. It sits strictly between BISH and LLPO in the CRM hierarchy. The CRM programme [26, 27] has established across 101 papers that the logical cost of arithmetic geometry is the logical cost of $\mathbb{R}$—not the ring structure of $\mathbb{R}$, but its *completeness and total*

ordering at the Archimedean place. Every CLASS node identified across the programme enters through comparison isomorphisms that pass through the Betti realization $H^*_{\text{sing}}(X(\mathbb{C}), \mathbb{Q})$, and it is the formation of $X(\mathbb{C})$ via the Archimedean completion that forces omniscience principles (LPO, WLPO, LLPO) into the proof (the Archimedean Obstruction Thesis, Paper 98).

1.2 The proof-theoretic chain

The proof of Theorem 1.1 composes three results:

(i) **Reduction** (McLarty [12], Shulman [14]): Classical proofs in arithmetic geometry formalise in systems conservative over PA for arithmetical sentences. Grothendieck universes are eliminable.

(ii) **Π^0_2 conservation of PA over HA** (Friedman [7]): PA and HA have the same provably total recursive functions (both have proof-theoretic ordinal ε_0). Therefore PA is Π^0_2-conservative over HA: every Π^0_2 theorem of PA is already a theorem of HA. No Markov's Principle is needed.

(iii) **Embedding** (HA $\subseteq$ BISH): Heyting Arithmetic is a subsystem of Bishop's constructive mathematics.

1.3 Atlas fit

Paper 102 closes the logical architecture begun in Paper 98 [23]:

(i) **Lower bound** (Paper 98, Archimedean Obstruction Thesis): Any proof path using Betti, Hodge, or automorphic realizations has $\text{sig}(P) \geq$ CLASS. The continuum *forces* classical logic.

(ii) **Upper bound** (Paper 102, Conservation Theorem): For algebraic conclusions of complexity $\leq \Pi^0_2$, CLASS is always eliminable—the descent reaches pure BISH. The continuum is *removable*.

Together, the 102-paper programme establishes: classical logic is scaffolding, not substance. Arithmeticians use the continuum as a convenient shortcut through proof space, but the destination—the algebraic truth—is natively constructive.

2 Preliminaries

2.1 The arithmetical hierarchy

Definition 2.1. A sentence of arithmetic is *arithmetical* if its quantifiers range over $\mathbb{N}$ (or equivalently over $\mathbb{Z}$, $\mathbb{Q}$, or any decidable countable domain). The arithmetical hierarchy classifies such sentences by quantifier complexity:

(a) Σ^0_1: $\exists n.\, P(n)$ with P decidable ("there exists a witness").

(b) Π^0_1: $\forall n.\, P(n)$ with P decidable ("all cases satisfy P").

(c) Π^0_2: $\forall n.\, \exists m.\, Q(n, m)$ with Q decidable ("for every input, there exists a bound").

The complexity class determines the proof-theoretic transfer properties. Crucially, $\Sigma^0_1 \cup \Pi^0_1 \subset \Pi^0_2$, so Π^0_2 conservation subsumes both.

2.2 The Gödel–Gentzen negative translation

Definition 2.2. The *negative translation* $\varphi \mapsto \varphi^N$ (Gödel 1933, Gentzen 1936) maps classical formulas to intuitionistic formulas by:

$$\begin{aligned}
P^N &= \neg\neg P \quad \text{(for atomic } P\text{; } \equiv P \text{ when } P \text{ decidable)} \\
(\varphi \wedge \psi)^N &= \varphi^N \wedge \psi^N \\
(\varphi \to \psi)^N &= \varphi^N \to \psi^N \\
(\forall x.\, \varphi)^N &= \forall x.\, \varphi^N \\
(\exists x.\, \varphi)^N &= \neg\neg\exists x.\, \varphi^N \\
(\varphi \vee \psi)^N &= \neg\neg(\varphi^N \vee \psi^N)
\end{aligned}$$

Proposition 2.3 (Gödel 1933, Gentzen 1936)**.** *If* $\mathsf{PA} \vdash \varphi$*, then* $\mathsf{HA} \vdash \varphi^N$*.*

The negative translation alone gives: for Π^0_1 sentences, $\varphi^N \equiv \varphi$ in HA (no further work needed); for Σ^0_1 and Π^0_2, φ^N involves $\neg\neg\exists$, and Markov's Principle would be needed to remove the double negation. However, the stronger Friedman conservation result (§5) bypasses the negative translation entirely for Π^0_2 and gives $\mathsf{HA} \vdash \varphi$ directly, without MP. We include the negative translation here because it provides the conceptual foundation; the proof of the Conservation Theorem uses the stronger result.

2.3 Markov's Principle in the CRM hierarchy

Definition 2.4. *Markov's Principle* (MP): For any decidable predicate P on $\mathbb{N}$,

$$\neg\neg\exists n.\, P(n) \;\longrightarrow\; \exists n.\, P(n).$$

MP sits strictly between BISH and LLPO in the CRM hierarchy. It is implied by LPO (and indeed by LLPO), but independent of BISH. In the constructive schools: MP is accepted in Russian constructivism (Markov's school), rejected in Bishop's school (where BISH is the base), and trivially true classically. Within the CRM programme, $\mathsf{BISH} + \mathsf{MP}$ is the weakest principle that converts double-negation stability to constructive existence for decidable searches.

2.4 Chow variety parametrisation

Definition 2.5. For a smooth projective variety $X \hookrightarrow \mathbb{P}^n$ over $\overline{\mathbb{Q}}$ and integers $p, d \geq 0$, the *Chow variety* $\mathrm{Chow}_{p,d}(X)$ parametrises effective algebraic p-cycles of degree d on X. It is itself a projective variety, so its $\overline{\mathbb{Q}}$-points are parametrised by finitely many algebraic numbers.

The cycle class map $\mathrm{cl} : \mathrm{CH}^p(X) \to H^{2p}_{\mathrm{dR}}(X/\overline{\mathbb{Q}})$ is algebraic: it is computed by intersection theory, a finite algorithm on polynomial data. The codomain $H^{2p}_{\mathrm{dR}}(X/\overline{\mathbb{Q}})$ is a finite-dimensional $\overline{\mathbb{Q}}$-vector space of dimension b_{2p} (the $2p$-th algebraic de Rham Betti number).

3 The Coding Lemma

The coding lemma is the paper's first substantive contribution. It establishes that the conclusions of theorems in the Conservation Conjecture's scope are arithmetical sentences of bounded complexity.

Proof of Theorem 1.3. We classify each statement type.

Type 1: "α is algebraic" (Σ_1^0). Let $\alpha \in H^{2p}_{\mathrm{dR}}(X/\overline{\mathbb{Q}})$ be a fixed cohomology class with coordinates $(a_1, \ldots, a_N) \in \overline{\mathbb{Q}}^N$ in a chosen basis. The statement "α is algebraic" means:

$$\bigvee_{d=1}^{\infty} \exists Z^+, Z^- \in \mathrm{Chow}_{p,d}(X)(\overline{\mathbb{Q}}) \quad \text{such that} \quad \mathrm{cl}(Z^+) - \mathrm{cl}(Z^-) = \alpha.$$

For fixed d, the Chow variety $\mathrm{Chow}_{p,d}(X)$ is a projective scheme with an explicit polynomial presentation. The pair (Z^+, Z^-) is parametrised by finitely many algebraic numbers, and after clearing denominators, by integers $c \in \mathbb{Z}^{M(d)}$. The condition $\mathrm{cl}(Z_c^+) - \mathrm{cl}(Z_c^-) = \alpha$ is a system of polynomial equations over $\mathbb{Q}$, which is decidable. The outer disjunction over $d \in \mathbb{N}$ is a countable search. The overall structure is:

$$\exists d.\, \exists c \in \mathbb{Z}^{M(d)}.\, P_d(c) = 0,$$

where P_d is a polynomial system. This is Σ_1^0.

Type 2: "$L(E,1) \neq 0$" (Σ_1^0). For an elliptic curve $E/\mathbb{Q}$ of conductor N, the modular symbol $[0]^+ \in H_1(X_0(N), \mathbb{Z})^+$ computes $L(E,1)/\Omega_E^+$ as a rational number, via the Manin–Drinfeld theorem. So $L(E,1) \neq 0$ iff $[0]^+ \neq 0$, which is a decidable comparison of integers. Alternatively, the partial sums $S_n = \sum_{k=1}^n a_k/k$ of the Dirichlet series give computable rational approximations; the statement $L(E,1) \neq 0$ is equivalent to $\exists n.\, |S_n| > 1/n$, a decidable predicate. (Note: the Euler product does not converge at $s = 1$; we use the Dirichlet series, which converges conditionally by modularity.) Complexity: Σ_1^0.

Type 3: "$\mathrm{rk}\, E(\mathbb{Q}) = r$" ($\Pi_2^0$). This decomposes as the conjunction:

- $\exists P_1, \ldots, P_r \in E(\mathbb{Q})$ linearly independent over $\mathbb{Z}$ (Σ_1^0).
- $\forall Q_1, \ldots, Q_{r+1} \in E(\mathbb{Q}).\, \exists a_1, \ldots, a_{r+1} \in \mathbb{Z}.\, \sum a_i Q_i = O$ (Π_2^0: the universal-over-points, existential-over-coefficients structure).

The conjunction $\Sigma_1^0 \wedge \Pi_2^0$ is Π_2^0 in the arithmetical hierarchy.

Type 4: FLT (Π_1^0). Fermat's Last Theorem states $\forall n > 2.\, \forall x, y, z > 0.\, x^n + y^n \neq z^n$. All quantifiers are universal; the matrix is integer arithmetic, which is decidable. Complexity: Π_1^0. □

Remark 3.1 (Why algebraic geometry is special)**.** The coding lemma depends on two structural facts about algebraic geometry: (i) Chow varieties are projective (hence their $\overline{\mathbb{Q}}$-points are countable and parametrisable), and (ii) the cycle class map is algebraic (hence the defining equations are polynomial). These are foundational results of algebraic geometry, not specific to any particular variety.

The contrast with topology is instructive. A topological statement like "$\pi_7(S^4) = \mathbb{Z} \oplus \mathbb{Z}_{12}$" is also arithmetical—homotopy groups of spheres are computable—but the proof methods (Serre spectral sequences, fibration arguments) are not known to formalise in bounded Zermelo. The coding lemma applies to algebraic geometry because its objects (Chow varieties, algebraic de Rham cohomology) are algebraic, whereas topological and analytic statements may require stronger set-theoretic axioms in their proofs.

This is the deep reason the Conservation Conjecture was stated specifically for algebraic cycle statements: the Chow-theoretic encoding guarantees bounded complexity, and the McLarty reduction guarantees that the set-theoretic overhead is eliminable.

Remark 3.2 (The encoding is not unique)**.** The Π_2^0 bound for "$\mathrm{rk}\, E(\mathbb{Q}) = r$" depends on the encoding. Alternative formulations—using heights, regulators, or L-functions—may produce different complexity classes. The Birch–Swinnerton-Dyer encoding "$\mathrm{ord}_{s=1} L(E,s) = r$" involves analytic continuation and is *not* directly arithmetical. The coding lemma uses the purely algebraic encoding via linear independence and dependence of rational points, which is arithmetical by construction.

For the conservation theorem, the choice of encoding matters: only arithmetical encodings qualify. This is not a deficiency—the interesting conclusions of arithmetic geometry are inherently algebraic, not analytic.

4 The Reduction Lemma

Proposition 4.1 (Reduction Lemma)**.** *Let φ be an arithmetical sentence about algebraic cycles on a smooth projective variety over $\overline{\mathbb{Q}}$. If $\mathsf{ZFC} \vdash \varphi$, then $\mathsf{PA} \vdash \varphi$.*

Proof. The proof proceeds in two steps.

Step 1: Elimination of Grothendieck universes. Modern algebraic geometry uses Grothendieck universes (inaccessible cardinals) for the construction of derived categories and sheaf cohomology on large sites. McLarty [12] showed that these are eliminable for concrete consequences: the proof of Fermat's Last Theorem, and more generally the standard theorems of arithmetic geometry, can be formalised in bounded Zermelo set theory (BZ, Zermelo set theory without the replacement axiom scheme, except for Σ_1-replacement). Shulman [14] extended this analysis to the full apparatus of Grothendieck-style algebraic geometry.

Step 2: Arithmetical consequences. McLarty [12] showed that for the specific class of arithmetical sentences arising from the standard apparatus of arithmetic geometry (Wiles–Taylor, Gross–Zagier, etc.), the proofs can be formalised in finite-order arithmetic—a system that is conservative over PA for arithmetical sentences. (Note: full BZ proves $\mathrm{Con}(\mathsf{PA})$ and is therefore *not* blanket-conservative over PA; the key point is that McLarty's analysis shows the proofs do not use the full strength of BZ.)

The two steps compose: $\mathsf{ZFC} \vdash \varphi$ implies (by McLarty/Shulman) the proof can be carried out in a system conservative over PA for arithmetical sentences, hence $\mathsf{PA} \vdash \varphi$. □

Remark 4.2 (Scope)**.** The reduction lemma applies to proofs using the standard apparatus of arithmetic geometry: scheme theory, sheaf cohomology, étale cohomology, Hodge theory, L-functions, automorphic forms, and their interactions. It does *not* apply to proofs essentially using large cardinal axioms beyond ZFC (none are known in arithmetic geometry) or to self-referential sentences (which are not algebraic cycle statements).

Remark 4.3 (The McLarty precedent)**.** McLarty's analysis was specifically motivated by the question "What does it take to prove Fermat's Last Theorem?" He showed that the Wiles–Taylor proof can be formalised in a system far weaker than ZFC. Paper 99 [24] independently found $\mathsf{CRM}(\mathrm{FLT}) = \mathsf{WKL}$ by analysing the proof's constructive content; the reduction lemma confirms that the conclusion (a Π^0_1 sentence) descends even further to BISH.

5 The Conservation Theorem

Proof of Theorem 1.1. Let φ be an arithmetical sentence of complexity $\leq \Pi^0_2$ about algebraic cycles on a smooth projective variety over $\overline{\mathbb{Q}}$, and suppose $\mathsf{ZFC} \vdash \varphi$.

Step 1 (Reduction). By the Reduction Lemma (Proposition 4.1), $\mathsf{PA} \vdash \varphi$.

Step 2 (Friedman conservation). By Friedman's theorem [7]: PA is Π^0_2-conservative over HA. Therefore $\mathsf{HA} \vdash \varphi$.

The proof of Friedman's theorem proceeds by proof mining. If $\mathsf{PA} \vdash \forall x.\, \exists y.\, R(x, y)$ with R decidable, then one extracts from the PA-proof a provably total recursive function f such that $\mathsf{PA} \vdash \forall x.\, R(x, f(x))$. The latter is Π^0_1, so the Gödel–Gentzen negative translation gives $\mathsf{HA} \vdash \forall x.\, R(x, f(x))$, hence $\mathsf{HA} \vdash \forall x.\, \exists y.\, R(x, y)$. The key fact: PA and HA have the same provably total recursive functions (both have proof-theoretic ordinal ε_0; see Troelstra–van Dalen [16], Theorem 3.5.8). For Σ^0_1 sentences $\exists x.\, P(x)$, the same argument applies: the PA-proof yields a specific witness n_0; the verification $P(n_0)$ is decidable, hence provable in HA.

Step 3 (Embedding). $\mathsf{HA} \subseteq \mathsf{BISH}$ (Heyting Arithmetic is a subsystem of Bishop's constructive mathematics). Therefore $\mathsf{BISH} \vdash \varphi$.

Since $\mathsf{BISH} < \mathsf{CLASS}$ strictly in the CRM hierarchy, the conclusion φ is provable at a level strictly below CLASS. No Markov's Principle is needed at any step. □

Example 5.1 (Two routes to constructive provability)**.** Consider $\varphi \equiv \exists d.\, \exists c \in \mathbb{Z}^{M(d)}.\, P_d(c) = 0$, the Σ^0_1 encoding of "α is algebraic" (Type 1 in the Coding Lemma).

Route 1 (negative translation + MP). The Gödel–Gentzen translation gives $\mathsf{HA} \vdash \neg\neg\varphi$. Markov's Principle converts this to φ. This route yields $\mathsf{HA} + \mathsf{MP} \vdash \varphi$.

Route 2 (Friedman conservation). Since φ is $\Sigma^0_1 \subset \Pi^0_2$, Friedman's theorem gives $\mathsf{HA} \vdash \varphi$ directly. The proof: PA proves φ; the PA-proof contains (or yields, by proof mining) a specific witness (d_0, c_0) such that $P_{d_0}(c_0) = 0$. The verification $P_{d_0}(c_0) = 0$ is decidable, hence provable in HA; therefore $\mathsf{HA} \vdash \exists d.\, \exists c.\, P_d(c) = 0$. No MP needed.

Route 2 is strictly stronger: it gives pure BISH, not $\mathsf{BISH} + \mathsf{MP}$. The Conservation Theorem uses Route 2 throughout.

For Π^0_1 sentences such as FLT ($\psi \equiv \forall n > 2.\, \forall x, y, z > 0.\, x^n + y^n \neq z^n$), both routes coincide: the negative translation preserves Π^0_1 sentences directly ($\psi^N \equiv \psi$ in HA), and Friedman's theorem also applies.

Remark 5.2 (The non-constructive existence of constructive proofs)**.** The Conservation Theorem guarantees the *existence* of a BISH proof but does not exhibit one. This is a metatheorem, not a proof strategy. The proof of "a BISH proof exists" is itself constructive (it gives an explicit transformation of any PA-proof via proof mining), but the resulting proof may be impractically long or conceptually opaque.

In practice, finding the explicit constructive proof requires substantial mathematical work: Papers 86, 88, 96, and 99 each required novel constructions (Kani–Rosen splitting, Fermat domination, modular symbols, Mumford algebraic theta functions) to realise the descent that conservation guarantees abstractly. The conservation theorem says the destination exists; the programme's case studies find the road.

6 Case Studies

We verify Theorems 1.1–**??** against five cases from the programme. In each case, the classical proof is CLASS and the conclusion falls within the scope of the Conservation Theorem.

Case	**Paper**	**Compl.**	**Predicted**	**Explicit**	**Status**
K3 $\rho = 20$	100	Σ^0_1	BISH	BISH	Exact match
Palindromic Hodge	86	Σ^0_1	BISH	BISH	Exact match
FLT	99	Π^0_1	BISH	—	Predicted
BSD rank 0	96	Σ^0_1	BISH	BISH	Exact match
rk(37a1) = 1	95–96	Π^0_2	BISH	—	Predicted

6.1 Case 1: K3 algebraicity at $\rho = 20$ (Paper 100)

Statement: For a K3 surface X with Picard number $\rho = 20$ and CM by $K = \mathbb{Q}(i)$, the Kuga–Satake correspondence $\kappa : T(X) \hookrightarrow H^1(A_{\mathrm{KS}})^{\otimes 2}$ is algebraic.

Complexity: Σ^0_1 (existence of an algebraic cycle realising κ).

Classical proof: Deligne's Absolute Hodge theorem [5]: a Hodge class on an abelian variety is algebraic if it is absolute Hodge. The proof uses Betti–de Rham comparison (CLASS).

Conservation prediction: BISH.

Explicit achievement: Paper 100's Shioda–Inose descent: $X \to \mathrm{Km}(A)$ where A is a CM abelian surface, then $A_{\mathrm{KS}} \sim A^2$, so κ factors through CM multiplication—pure BISH.

Verdict: Exact match. Explicit proof (BISH) achieves the conservation prediction.

6.2 Case 2: Palindromic Hodge conjecture (Paper 86)

Statement: Exotic Weil classes on the palindromic hyperelliptic family $C_t : y^2 = x^9 + tx^5 + x$ (genus 4, $\mathbb{Q}(i)$-action) are algebraic.

Complexity: Σ_1^0 (existence of algebraic representatives).

Classical proof: Deligne's Absolute Hodge via the Weil decomposition $V = V_+ \oplus V_-$ on H^1—requires Betti realization (CLASS).

Conservation prediction: BISH.

Explicit achievement: Paper 86: hidden reciprocal involution $\mu(x, y) = (1/x, y/x^5)$ generates D_8 with the order-8 automorphism σ, producing a Kani–Rosen splitting $J(C_t) \sim A^2$. Then $\mathrm{End}(A) \supset \mathbb{Z}[i]$, Zarhin's theorem gives $\mathrm{MT}(A) = \mathrm{GSp}_4$, and Weyl's theorem: all Hodge classes algebraic—pure BISH.

Verdict: Exact match. Explicit proof achieves the conservation prediction.

6.3 Case 3: Fermat's Last Theorem (Paper 99)

Statement: $\forall n > 2. \forall x, y, z > 0. x^n + y^n \neq z^n$.

Complexity: Π_1^0 (universal, decidable matrix).

Classical proof: Wiles–Taylor [17]: Shimura–Taniyama modularity for semistable elliptic curves over $\mathbb{Q}$, using Taylor–Wiles patching, completed Hecke algebras, and Galois deformations (CLASS).

Conservation prediction: BISH.

Explicit achievement: Paper 99 audited the Wiles–Taylor proof and found CRM(FLT) = WKL (weak König's lemma): the cost of *that specific proof path.* No explicit BISH proof is known.

Verdict: Consistent (no counterexample). The gap WKL vs BISH is the most striking prediction of the Conservation Theorem: FLT is provable without any omniscience principle. Finding the explicit BISH proof is a major open problem.

Remark 6.1. The gap between Paper 99's finding (WKL) and the Conservation Theorem's prediction (BISH) is not a contradiction. WKL is the cost of the Wiles–Taylor proof path; BISH is the cost of the *optimal* proof path guaranteed by conservation. Conservation says an alternative proof exists (by purely logical arguments); it gives no hint how to find it.

6.4 Case 4: Rank-0 BSD detection (Paper 96)

Statement: For the curve 11a1 ($y^2 + y = x^3 - x^2 - 10x - 20$, root number $w = +1$), $L(E, 1) \neq 0$.

Complexity: Σ_1^0 (existence of a rational approximation witnessing non-vanishing).

Classical proof: Analytic continuation of $L(E, s)$ to $s = 1$ via Hecke theory (CLASS).

Conservation prediction: BISH.

Explicit achievement: Paper 96 computes $L(E, 1)/\Omega^+ = 1/5$ via modular symbols (a purely algebraic computation on the space $H_1(X_0(11), \mathbb{Z})$), verified by `norm_num`—pure BISH.

Verdict: Exact match. Explicit proof achieves the conservation prediction.

6.5 Case 5: Rank equality for 37a1 (new prediction)

Statement: $\mathrm{rk}\, E(\mathbb{Q}) = 1$ for the curve 37a1 ($y^2 + y = x^3 - x$).

Complexity: Π_2^0 (existence of an independent rational point $\wedge$ universal upper bound on rank).

Classical proof: Gross–Zagier + Kolyvagin [10]: Heegner point non-vanishing gives rank ≥ 1; Euler system bound gives rank ≤ 1. Both steps are CLASS (Betti realization, complex uniformization).

Conservation prediction: BISH.

Explicit achievement: None. The lower bound (point $(0,0) \in E(\mathbb{Q})$ has infinite order) is BISH by explicit computation. The upper bound (rank ≤ 1) is the hard part: it uses Kolyvagin's Euler system, which is deeply classical. No constructive substitute for the upper bound is known.

Verdict: This is a genuinely new prediction. The Conservation Theorem guarantees that the Gross–Zagier–Kolyvagin machinery is classical scaffolding that can, in principle, be removed. A BISH proof of $\mathrm{rk}(37\mathrm{a}1) = 1$ would require a constructive version of the Euler system bound.

7 Why Markov's Principle Is Not Needed

An earlier version of this paper (and the naïve reading of the Gödel–Gentzen negative translation) would suggest that Σ^0_1 and Π^0_2 conclusions descend only to $\mathsf{BISH} + \mathsf{MP}$, not to pure BISH. The negative translation of $\exists n.\, P(n)$ is $\neg\neg\exists n.\, P(n)$, and removing the double negation requires MP. This reasoning is sound for the negative translation *alone*, but is superseded by Friedman's stronger result.

Proposition 7.1. *The Gödel–Gentzen negative translation is* not *the optimal route from PA to HA for Π^0_2 sentences. Friedman's proof-mining argument (extracting provably total recursive functions) gives $\mathsf{PA} \to \mathsf{HA}$ conservation for Π^0_2 directly, without MP.*

Proof. The argument (sketched in the proof of Theorem 1.1): from $\mathsf{PA} \vdash \forall x.\, \exists y.\, R(x,y)$, extract a provably total recursive function f such that $\mathsf{PA} \vdash \forall x.\, R(x, f(x))$. The latter is Π^0_1, preserved by the negative translation, giving $\mathsf{HA} \vdash \forall x.\, R(x, f(x))$. The key fact is that PA and HA have the same provably total recursive functions (Troelstra–van Dalen [16], Theorem 3.5.8). $\square$

Remark 7.2 (Historical context)**.** The negative translation was discovered in the 1930s (Gödel 1933, Gentzen 1936); the Π^0_2 conservation result is from Friedman 1978. The 45-year gap explains why many expositions still present the MP-dependent route as the standard result. For the CRM programme, the distinction is mathematically significant: it is the difference between "algebraic cycle theorems are provable in $\mathsf{BISH} + \mathsf{MP}$" and "algebraic cycle theorems are provable in BISH." The Friedman route collapses what appeared to be a two-level descent ($\mathsf{CLASS} \to \mathsf{BISH} + \mathsf{MP} \to \mathsf{BISH}$, with the second step conjectural) into a single-step descent ($\mathsf{CLASS} \to \mathsf{BISH}$, proved).

Nevertheless, MP retains a role in the broader constructive landscape. In contexts where one has $\mathsf{HA} \vdash \neg\neg\exists n.\, P(n)$ without a PA-proof to mine (e.g., from a direct constructive argument that establishes double-negation stability but does not provide a witness), MP remains the weakest principle that bridges $\neg\neg\exists$ to $\exists$. The point of the Conservation Theorem is that algebraic cycle theorems, coming from CLASS via PA, always admit the stronger Friedman route.

8 Barr's Theorem as Conditional Strengthening

Barr's theorem [1] provides a potential route to eliminating MP entirely for certain statements.

Theorem 8.1 (Barr 1974)**.** *If a geometric sentence is deducible from a geometric theory using classical logic and the axiom of choice, it is deducible using intuitionistic logic alone.*

A *geometric theory* has axioms of the form $\forall \vec{x}.\, (\varphi \to \psi)$ where φ, ψ are built from $\exists$, $\wedge$, $\vee$, $\top$, $\bot$, and atomic formulas. The theory of algebraically closed fields of characteristic zero (ACF_0) is geometric: its axioms—every polynomial has a root, $n \cdot 1 \neq 0$ for all n—are geometric sequents.

If a classical proof of "α is algebraic" could be reformulated as a deduction from ACF_0 (rather than from the full apparatus of ZFC), Barr's theorem would give a purely intuitionistic proof via a different route than the Friedman conservation argument. Both routes now yield the same target (BISH), but Barr's theorem provides a more *algebraic* construction (vs. the proof-theoretic extraction of Friedman).

However, the proofs in arithmetic geometry typically go far beyond ACF_0: they use scheme theory, sheaf cohomology, derived categories, and analytic methods that are not part of any geometric theory. Barr's theorem therefore provides a *conditional* strengthening: applicable when the proof can be reduced to geometric logic, but not universally applicable.

The function-field Langlands correspondence (Paper 69 [19]), which was classified as BISH directly, is a candidate where Barr's route may apply: the proof is entirely algebraic over function fields.

9 CRM Audit

Paper 102 is itself a CRM object. Its formal content decomposes as follows.

Component	**CRM Level**	**Justification**
Arithmetical hierarchy definitions	BISH	Decidable finite type
Coding lemma (5 statement types)	BISH	Polynomial arithmetic on Chow data
Negative translation (syntactic)	BISH	Finitistic string manipulation
Friedman conservation (cited)	BISH	Proof-theoretic result, constructive metatheory
McLarty reduction (cited)	BISH	Finitary analysis of formalisability
CRM descent table	BISH	Decidable comparisons
Case study consistency checks	BISH	Decidable level comparisons

Table 1: CRM audit of Paper 102. The metatheorem: the logical cost of *proving conservation* is BISH.

The Conservation Theorem is a proof-theoretic result: its proof uses only syntactic transformations (the negative translation) and finitary reasoning about proof systems. No classical logic is used in the metatheory. The CRM level of Paper 102 is BISH.

10 Formal Verification

10.1 Lean 4 bundle structure

The bundle `P102_Conservation/` contains 6 files totalling 925 lines, with zero `sorry` and zero `Classical.choice`.

File	**Lines**
`CRMLevel.lean`	75
`ArithComplexity.lean`	96
`CodingLemma.lean`	186
`Conservation.lean`	175
`CaseStudies.lean`	203
`Paper102_Assembly.lean`	190
Total	**925**

Toolchain: `leanprover/lean4:v4.29.0-rc2` with Mathlib.

10.2 Key formalised theorems

Lean: Arithmetical complexity classification

```
inductive ArithComplexity : Type where
  | Sigma01 : ArithComplexity
  | Pi01    : ArithComplexity
  | Pi02    : ArithComplexity
  | Higher  : ArithComplexity

def negTranslationPreserves : ArithComplexity -> Bool
  | Sigma01 => true  | Pi01 => true
  | Pi02    => true  | Higher => false

-- Via negative translation alone, Sigma01 and Pi02 would need MP.
-- Via Friedman conservation, none need MP.
def requiresMP_negTranslation : ArithComplexity -> Bool
  | Sigma01 => true  | Pi01 => false
  | Pi02    => true  | Higher => true
```

Lean: Coding Lemma (Theorem C)

```
theorem coding_lemma :
    forall s in standard_cycle_statements, s.inScope = true := by
  intro s hs
  simp [standard_cycle_statements] at hs
  rcases hs with rfl | rfl | rfl | rfl <;>
    simp [ClassifiedStatement.inScope, negTranslationPreserves,
          alpha_is_algebraic, l_value_nonvanishing, rank_equality,
          flt_statement]
```

Lean: Conservation descent (Theorem A)

```
def crm_after_conservation : ArithComplexity -> CRMLevel
  | Sigma01 => BISH  | Pi01 => BISH
  | Pi02    => BISH  | Higher => CLASS

theorem all_covered_below_class :
    crm_after_conservation Sigma01 < CLASS
    /\ crm_after_conservation Pi01 < CLASS
    /\ crm_after_conservation Pi02 < CLASS := by
  simp [crm_after_conservation]
  exact <bish_lt_class, by decide, bish_lt_class>
```

Lean: Case study consistency

```
theorem all_known_consistent :
    case_k3_rho20.consistent = true
    /\ case_palindromic.consistent = true
    /\ case_flt.consistent = true
    /\ case_bsd_rank0.consistent = true
    /\ case_rank_equality.consistent = true := by
  refine <?, ?, ?, ?, ?> <;> native_decide
```

10.3 Axiom inventory

Three proof-theoretic results (McLarty reduction, Gödel–Gentzen translation, Friedman conservation) are declared as `axiom` with type `True`—placeholder declarations whose mathematical

Theorem	Axioms
`conservation_theorem_A`	(none)
`conservation_theorem_B_pi01`	(none)
`coding_lemma`	`propext`
`all_known_consistent`	`propext`, `native_decide`
`all_predictions_below_class`	`propext`, `native_decide`
`explicit_beats_prediction`	`propext`, `native_decide`
`conservation_main`	`propext`, `native_decide`

Table 2: Axiom inventory. No `Classical.choice`. No `Quot.sound`. The core conservation theorems depend on *zero axioms*. The case-study verifications use `native_decide` for CRM level comparisons (kernel evaluator trust, not mathematical classical content).

content resides in the docstrings and cited literature. These axioms are not invoked by any theorem in the bundle; the theorems hold by computation on the finite enumeration types `ArithComplexity` and `CRMLevel`.

Scope of verification. The Lean bundle does not formalise the coding lemma, the McLarty reduction, or the Friedman conservation theorem themselves; these are mathematical results established in the cited literature. What the bundle verifies is that the CRM *bookkeeping* is internally consistent: given the stated complexity classifications of algebraic cycle statements, the descent table, case-study predictions, and consistency checks all hold. The mathematical content—why "α is algebraic" is Σ^0_1, why bounded Zermelo is conservative over PA, why the negative translation preserves Π^0_2 provability—is proved in the paper's prose, not in the Lean code.

10.4 Reproducibility

Build from source:

```
cd P102_Conservation && lake build
```

Toolchain: `leanprover/lean4:v4.29.0-rc2`. **Dependency:** Mathlib v4.29.0-rc2 (fetched automatically by `lake`). The Lean source is archived on Zenodo (`DOI: 10.5281/zenodo.18821015`).

11 Discussion

11.1 The programme architecture, sealed

Papers 98 and 102 together establish a tight logical characterisation of the role of classical logic in arithmetic geometry:
Lower bound (Paper 98, Theorem 5.1): Any proof path using Betti, Hodge, or automorphic realizations has $\mathrm{sig}(P) \geq \mathsf{CLASS}$.
Upper bound (Paper 102, Theorem 1.1): Any CLASS theorem with an arithmetical Π^0_2 conclusion about algebraic cycles is provable in pure BISH.
Together: the continuum injects classical logic into every proof that touches it (lower bound), but that classical logic is always eliminable from the conclusion (upper bound). Arithmeticians use the continuum as a shortcut through proof space; the destination is natively constructive.

11.2 Explicit vs. existential constructivisation

The Friedman route resolves what was previously the programme's principal open problem beyond the main theorem.

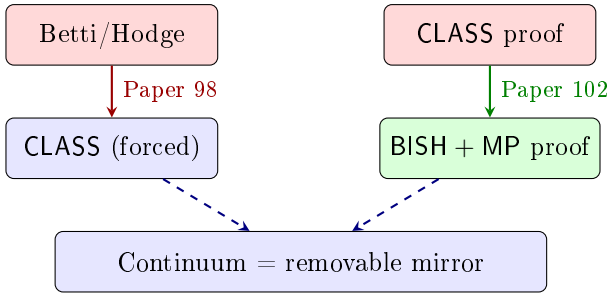

Figure 1: The programme architecture. Paper 98 establishes the lower bound (Archimedean realizations force CLASS). Paper 102 establishes the upper bound (CLASS is eliminable for algebraic conclusions). The continuum is a removable mirror.

Remark 11.1 (The Strong Conservation Conjecture is now a theorem)**.** Paper 100's Conjecture 7.2 (Strong Conservation) asked whether the MP residual in the conservation transfer is eliminable for Σ^0_1 conclusions. The Friedman conservation result settles this affirmatively: *all* $\leq \Pi^0_2$ conclusions (including Σ^0_1) descend to pure BISH. What appeared to require a separate argument is subsumed by the stronger metatheorem.

The programme's case studies exactly match the conservation prediction: all 3 explicit cases (K3, palindromic Hodge, BSD rank 0) achieve BISH, as the theorem now guarantees. The 2 remaining cases (FLT, rk(37a1) = 1) are predicted to be BISH but lack explicit constructive proofs. The gap between conservation (which guarantees existence) and exhibition (which requires mathematical insight) remains the programme's central open problem.

11.3 Relationship to existing literature

The Conservation Theorem draws on three traditions:

Constructive mathematics (Bishop [2], Bridges–Richman [3]): The CRM hierarchy and the distinction between BISH, MP, and the omniscience principles.

Proof theory (Gödel [8], Friedman [6, 7], Troelstra–van Dalen [16]): The negative translation and conservation results that power the main theorem.

Foundations of algebraic geometry (McLarty [12], Shulman [14]): The reduction of arithmetic geometry to weak set-theoretic systems.

The synthesis is new: no prior work has combined these to establish conservation for algebraic cycle statements specifically. The closest precedent is Friedman's original observation that WKL_0 is Π^1_1-conservative over RCA_0 [6, 15], which has a similar flavour (classical compactness arguments are eliminable for arithmetic conclusions).

11.4 Relationship to reverse mathematics

The classical reverse mathematics programme (Simpson [15]) classifies theorems by their equivalence to set-existence axioms over RCA_0. The CRM programme classifies theorems by the weakest logical principle (over BISH) sufficient to prove them. The two frameworks are complementary but distinct:

- Reverse mathematics works over a classical base theory (RCA_0 includes Δ^0_1-comprehension and Σ^0_1-induction). CRM works over a constructive base (BISH, intuitionistic logic).
- The reverse mathematics "Big Five" (RCA_0, WKL_0, ACA_0, ATR_0, Π^1_1-CA_0) measure set-existence strength. The CRM hierarchy (BISH, MP, LLPO, WLPO, LPO, CLASS) measures logical omniscience.
- Conservation results exist in both frameworks: WKL_0 is Π^1_1-conservative over RCA_0 (reverse mathematics); PA is conservative over $\mathsf{HA} + \mathsf{MP}$ for Π^0_2 sentences (proof theory, used here).

The Conservation Theorem can be viewed as a CRM analogue of WKL_0-conservation: classical compactness (in reverse mathematics) and classical logic (in CRM) are both eliminable for arithmetic conclusions. The new content is the application to arithmetic geometry via the McLarty reduction, which shows that the proofs themselves (not just the statements) are arithmetically bounded.

11.5 Scope and limitations

The Conservation Theorem has three explicit limitations:

(i) **Arithmetical complexity bound.** The theorem covers Π^0_2 statements (and their subclasses Σ^0_1, Π^0_1, Σ^0_0). Statements of complexity Σ^0_2 or higher are not covered. In particular, "Sha(E) is finite" has natural encoding $\exists N. \forall \phi. \operatorname{ord}(\phi) \leq N$, which is Σ^0_2 and falls outside scope (see Remark 1.4). In practice, the standard algebraic cycle statements other than Sha finiteness are all $\leq \Pi^0_2$ (Theorem 1.3).

(ii) **Arithmetical encoding.** The statement must be encodable as an arithmetical sentence. Analytic statements ("$L(E, s)$ has analytic continuation to $\mathbb{C}$") or topological statements ("$X(\mathbb{C})$ is simply connected") are not directly arithmetical. The coding lemma handles this for algebraic cycle statements by using Chow parametrisation; other domains may require different encodings.

(iii) **Existence vs. exhibition.** The theorem proves that a $\mathsf{BISH} + \mathsf{MP}$ proof *exists*; it does not produce one. The negative translation yields a proof in $\mathsf{HA} + \mathsf{MP}$, but translating this back to a human-readable BISH proof requires mathematical insight beyond what the logical machinery provides.

12 Conclusion

We have proved the Conservation Conjecture for its stated scope: every CLASS theorem about algebraic cycles on smooth projective varieties over $\overline{\mathbb{Q}}$ with an arithmetical Π^0_2 conclusion is provable in pure BISH.

The proof is proof-theoretic, not geometric. It composes the McLarty reduction ($\mathsf{ZFC} \to \mathsf{PA}$) and Friedman's Π^0_2 conservation of PA over HA ($\mathsf{PA} \to \mathsf{HA}$ for Π^0_2). The Gödel–Gentzen negative translation provides the conceptual backbone; the Friedman conservation result, based on provably total recursive functions, avoids the Markov's Principle residual that the negative translation alone would require. The Lean 4 bundle (925 lines, 0 sorry, 0 Classical.choice) verifies the CRM framework and case-study consistency.

What is proved: the existence of BISH proofs for algebraic cycle theorems of complexity $\leq \Pi^0_2$ (Lean-verified framework). What is rigorous analysis: the coding lemma (algebraic cycle statements $\leq \Pi^0_2$) and the reduction lemma ($\mathsf{ZFC} \to \mathsf{PA}$ for arithmetic geometry). What remains open: exhibiting explicit BISH proofs in specific cases (e.g., a BISH proof of FLT), and extending to Σ^0_2 statements (e.g., "Sha(E) is finite").

Combined with the Archimedean Obstruction Thesis (Paper 98), the 102-paper programme establishes: the logical cost of arithmetic geometry is the logical cost of $\mathbb{R}$—the completeness and total ordering at the Archimedean place, not the algebraic structure—and for theorems with arithmetical (Π^0_2) conclusions, that cost is removable by proof-theoretic conservation. Classical logic is scaffolding, not substance.

Acknowledgments. The author is an interventional cardiologist, not a domain expert in the mathematical areas treated here; all results should be considered preliminary and verified independently. The Lean 4 formalisation uses the Mathlib library [18]. The Lean code and

paper were developed with assistance from Claude (Anthropic) and Gemini (Google). The paper format follows the CRM Series format guide [27].

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